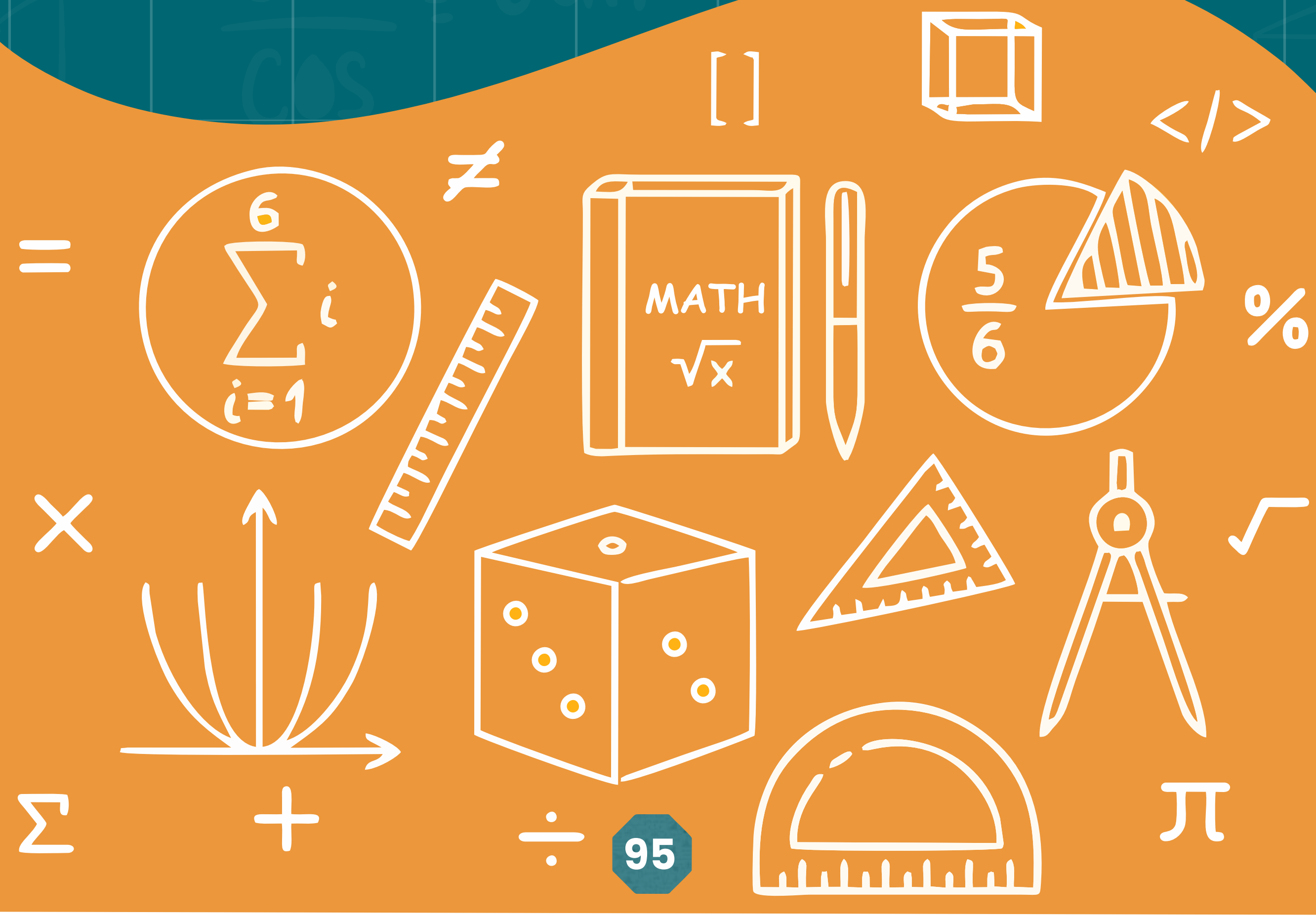
UNIT
9

LINEAR EQUATIONS



LINEAR EQUATIONS

A linear equation is an equation that has only one variable, and the power of the variable is 1.

Examples:

1. $2p + 4 = 8$
2. $5 - 3y = -7$
3. $(1/2)a - 4 = 6$

In all these examples:
• There is only **one variable** (like **p**, **y**, or **a**).
• The power of the variable is **1**.

Rules for Solving a Linear Equation

To solve a linear equation and find the value of the variable, we must follow these rules:

1. Add the same number to both sides of the equation.
Example: If $x - 3 = 7$, add 3 to both sides: $x - 3 + 3 = 7 + 3$, so $x = 10$.

2. Subtract the same number from both sides of the equation.
Example: If $x + 5 = 12$, subtract 5 from both sides: $x + 5 - 5 = 12 - 5$, so $x = 7$.

3. Multiply both sides of the equation by the same non zero number.
Example: If $x/2 = 6$, multiply both sides by 2: $(x/2) \times 2 = 6 \times 2$, so $x = 12$.

4. Divide both sides of the equation by the same non zero number.
Example: If $3x = 15$, divide both sides by 3: $(3x)/3 = 15/3$, so $x = 5$.

Steps to Solve a Linear Equation

1. Simplify both sides of the equation:
Remove brackets or parentheses and combine like terms.

2. Remove fractions:
Multiply both sides by the denominator to get rid of fractions.

3. Rearrange terms:
Move all terms with the variable to one side and constants to the other side.

4. Make the coefficient of the variable equal to 1:
Divide or multiply to isolate the variable.

Let's see some examples.

Example 1. Solve $3m + 5 = 25 - 2m$.
Solution. $3m + 5 = 25 - 2m$
$\Rightarrow 3m + 2m = 25 - 5$
$\Rightarrow 5m = 20$
$\Rightarrow m = 20 \div 5$
$\Rightarrow m = 4$

Example 2. Solve $2(p - 1) = p + 12$.
Solution. $2(p - 1) = p + 12$
$\Rightarrow 2p - 2 = p + 12$
$\Rightarrow 2p - p = 12 + 2$
$\Rightarrow p = 14$

Example 3. If 7 is added to three times of a number, then the result is 28. Find the number.
Solution. Let's assume the number is p.
As per the word problem given, linear equation will be
$3p + 7 = 28$
$\Rightarrow 3p = 28 - 7$
$\Rightarrow 3p = 21$
$\Rightarrow p = 21 \div 3$
$\Rightarrow p = 7$
Hence, the number is 7.

Example 4. Find the three consecutive odd numbers whose sum is 105.
Solution. Let's consider the smallest, odd number is equal to m.
Next two odd numbers are m+2 and m+4.
According to the given word problem following linear equation can be formed.
$m + m + 2 + m + 4 = 105$
$\Rightarrow 3m + 6 = 105$

$\Rightarrow 3m = 99$
$\Rightarrow m = 99 \div 3$
$\Rightarrow m = 33$

Hence, the required consecutive odd numbers are 33, 35 and 37.

5. Solving linear equations involving brackets.

Example 1. Solve: $8(x - 3) - (6 - 2x) = 2(x + 2) - 5(5 - x)$and verify.

Solution:

We begin by multiplying out the brackets, taking care, in particular, with any signs of terms.

$$8x - 24 - 6 + 2x = 2x + 4 - 25 + 5x$$

Each side can be tidied up by collecting the x terms and the numbers together.

$$8x + 2x - 24 - 6 = 2x + 5x + 4 - 25$$

or $\quad 10x - 30 = 7x - 21$

Now subtract 7x from both sides:

$$10x - 7x - 30 = \cancel{7x} - \cancel{7x} - 21$$

or $\quad 3x - 30 = -21$

Add 30 on both sides:

$$3x - \cancel{30} + \cancel{30} = 30 - 21$$

or $\quad 3x = 9$

or $\quad \dfrac{3x}{3} = \dfrac{9}{3} = 3$ (Dividing by 3 on both sides)

Therefore, x = 3.

Verification: Put x=3 in the original equation.

We have, $\quad 8(x - 3) - (6 - 2x) = 2(x + 2) - 5(5 - x)$

$$8(3 - 3) - (6 - 2 \times 3) = 2(3 + 2) - 5(5 - 3)$$
$$8(0) - (6 - 6) = 2(5) - 5(2)$$
$$0 - 0 = 10 - 10$$
$$0 = 0$$

Hence verified, x = 3

6. Solving linear equations with fractional coefficients

Example 1.

Solve: $\frac{4(x+2)}{5} = 7 + \frac{5x}{13}$

Multiplying both sides by the lowest common multiple of denominators, which is 5 × 13 = 65.

$$65 \times \frac{4(x+2)}{5} = 65\left(7 + \frac{5x}{13}\right)$$

or $\overset{13}{\cancel{65}} \times \frac{4(x+2)}{\cancel{5}_1} = 65 \times 7 + \overset{5}{\cancel{65}} \times \frac{5x}{\cancel{13}_1}$

or $52(x+2) = 455 + 25x$

or $52x + 104 = 455 + 25x$

Subtracting 25x from both sides:

or $52x - 25x + 104 = 455 + 25x - 25x$

or $27x + 104 = 455$

Subtracting 104 from both sides:

or $27x + 104 - 104 = 455 - 104$

or $27x = 351$

Dividing on both sides by 27:

Then $\frac{\cancel{27}x}{\cancel{27}} = \frac{\overset{13}{\cancel{351}}}{\cancel{27}_1}$

So $x = 13$

Example 2. Solve: $\frac{x+5}{6} - \frac{x+1}{9} = \frac{x+3}{4}$ and verify

Solution:

(LCM of 4, 6 and 9 is 36)

$\frac{\overset{6}{\cancel{36}}(x+5)}{\cancel{6}_1} - \frac{\overset{4}{\cancel{36}}(x+1)}{\cancel{9}_1} = \frac{\overset{9}{\cancel{36}}(x+3)}{4}$ (Multiplying both sides by LCM 36)

Therefore, $\frac{6(x + 5)}{1} - \frac{4(x + 1)}{1} = \frac{9(x + 3)}{1}$

or $= 6(x + 5) - 4(x + 1) = 9(x + 3)$
or $= 6x + 30 - 4x - 4 = 9x + 27$
or $= 6x - 4x + 30 - 4 = 9x + 27$
or $= 2x + 26 = 9x + 27$
or $= 26 = 9x + 27 - 2x$
or $= 26 = 7x + 27$
or $= 26 - 27 = 7x$
or $= -1 = 7x$
or $= 7x = -1$
or $= \frac{7x}{7} = \frac{-1}{7}$

Thus, $x = -\frac{1}{7}$

Verification:
Putting $x = -\frac{1}{7}$

$$\frac{x + 5}{6} - \frac{x + 1}{9} = \frac{x + 3}{4}$$

or $$\frac{-\frac{1}{7} + 5}{6} \quad \frac{-\frac{1}{7} + 1}{9} = \frac{-\frac{1}{7} + 3}{4}$$

$$\left(\frac{-1 + 35}{7}\right) \times \frac{1}{6} - \left(\frac{-1 + 7}{7}\right) \times \frac{1}{9} = \left(\frac{-1 + 21}{7}\right) \times \frac{1}{4}$$

$$\frac{\cancel{34}^{17}}{\cancel{42}_{21}} - \frac{\cancel{6}^{2}}{\cancel{63}_{21}} = \frac{20}{28}$$

$$\frac{17}{21} - \frac{2}{21} = \frac{\cancel{20}^{5}}{\cancel{28}_{7}}$$

or $$\frac{17 - 2}{21} = \frac{5}{7}$$

or $$\frac{\cancel{15}^{5}}{\cancel{21}_{7}} = \frac{5}{7}$$

or $\frac{5}{7} = \frac{5}{7}$

Hence verified, $x = -\frac{5}{7}$

EXERCISE 9.1

Q1. Solve the following linear equations.

1. $7x + 9 = 23$
2. $5x + 7 = 42$
3. $4x + 3 = 51$
4. $9x + 5 = 41$
5. $4x + 2 = 34$
6. $11x + 3 = 36$

Q2. Solve the following.

1) $\frac{f}{5} + 2 = 8$
2) $\frac{w}{3} - 5 = 2$
3) $\frac{x}{8} + 3 = 12$
4) $\frac{5f}{4} + 3 = 18$
5) $\frac{3y}{2} - 1 = 8$
6) $\frac{2x}{3} + 5 = 12$
7) $\frac{f}{5} + 3 = 1$
8) $\frac{x + 3}{2} = 5$
9) $\frac{t - 5}{2} = 3$

Q3. Solve Linear equations.

a) $2x + 10 = 7x + 5$
b) $3x + 9 = 6x$
c) $3(x + 4) = 6x$
d) $8y - 6 = 2(y + 9)$
e) $3(a + 4) = 5(a + 2)$

Q4. Solve and verify.

1) $\frac{5a}{3} - 2 = \frac{a}{4} + 15$
2) $\frac{x}{6} + \frac{x}{6} - 1 = \frac{5x}{12}$
3) $\frac{2a}{5} - \frac{7}{5} = \frac{3a}{4} - \frac{7}{5}$
4) $\frac{x}{2} - \frac{x}{3} = \frac{x}{4} - 1$
5) $\frac{3x}{4} + \frac{1}{2} = \frac{x}{5} + \frac{8}{5}$
6) $\frac{x}{4} - \frac{5}{12} = \frac{x}{5} - \frac{x}{3}$

7) $\frac{x}{5} = \frac{3}{35} + \frac{x+1}{7}$

8) $\frac{2y+1}{3} + 3 = y$

9) $\frac{5x+2}{3} + \frac{x}{5} = \frac{3x-5}{15} + x$

10) $\frac{1}{2} + \frac{x-1}{3} = \frac{x}{2}$

11) $x - \frac{x-1}{2} = 0$

12) $\frac{4x}{3} - \frac{3x-4}{6} = 5 - \frac{x-2}{2}$

Q5. Solve and verify the following equations:

1. $3x + 9 = 18$

2. $2x - 5 = 9$

3. $4x - 5 = 11$

4. $5x - 20 = 30$

5. $3x + (-17) = 2$

6. $13 - 3x = -2$

7. $4x + 8 = 2$

8. $\frac{2y-1}{3} + 3 = y$

9. $\frac{5x+2}{3} + \frac{x}{5} = \frac{3x-5}{15} + x$

10. $\frac{1}{2} + \frac{x-1}{3} = \frac{x}{2}$

11. $x - \frac{x-1}{2} = 0$

12. $\frac{4x}{3} - \frac{3x-4}{6} = 5 - \frac{x-2}{2}$

1. Convert the following statements into equations.
(a) 5 added to a number is 9.
(b) 3 subtracted from a number is equal to 12.
(c) 5 times a number decreased by 2 is 4.
(d) 2 times the sum of the number x and 7 is 13.

2. A number is 12 more than the other. Find the numbers if their sum is 48.

3. Twice the number decreased by 22 is 48. Find the number.

4. Seven times the number is 36 less than 10 times the number. Find the number.

WORD PROBLEMS

1. Aslam had Rs 35 in his pocket. He purchased three pencils. Still, he has Rs 5 in his pocket. Find the price of pencils.

2. Ali has some amount of rupees. If Rs 7 is added to thrice that amount, it makes Rs 22. What amount is with him?

3. The length of a room is 1.5m more than its breadth. If the perimeter of the room is 63m, find the length and breadth of the room.

4. The cost of 3 notebooks and 5 similar pens is Rs. 460. If the notebook's cost is Rs. 20 more than the pen's, find the cost of each.